

We Live as We Dream: The Cognitive Science of Coming Home

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Introduction to Cognitive Science

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“It is impossible to convey the life-sensation of any given epoch of one’s existence... its subtle and penetrating essence. It is impossible. We live, as we dream — alone.”

- Joseph Conrad, *Heart of Darkness*

Conrad expresses what feels like a poetic truth, and yet, through the proper lens, we can see it is actually a structural fact about how the mind works. Conrad poignantly identifies what feels like an insurmountable chasm between experience and expression; cognitive science explains this gap as the product of automatic, encapsulated systems that operate outside of our conscious awareness. The mind is not a blank slate but rather a collection of evolved, specialized systems, which the field calls cognitive architecture. Operating under the phenomenon known as instinct blindness, these modular systems for perception, reasoning, and language run automatically, inferring structure from the world without our conscious oversight. Together, they create a unified experience of the world until the environment shifts in ways that violate their built-in assumptions.

Combat environments excel at activating this architecture, while returning home forces these same systems into contexts they were not designed to process. In combat, subtle motion cues reliably signal danger. This results in tuning the animacy detection and vigilance towards a higher sensitivity. Failure to do so could be catastrophic. At home, this same vigilance generates false positives in crowded, ambiguous settings. Similarly, combat relies on explicit social rules in which the cheater-detection system thrives, whereas civilian life is governed by diffuse, unenforced norms. Even language, which functions as a life-saving shibboleth in the military, becomes a source of alienation when those signals no longer carry shared meaning. Because these systems are informationally encapsulated, their outputs cannot simply be overridden by conscious belief: knowing that one is safe in a grocery store does not prevent low-level threat

modules from firing automatically. What feels like a personal failing is often a structural mismatch between evolved architecture and modern input.

In this paper, I argue that cognitive science provides a robust framework for understanding this mismatch. Specifically, combat situations activate systems for threat detection, social bonding, and language in ways that closely resemble the ancestral conditions they evolved for, while modern civilian life violates their built-in assumptions. This evolutionary and architectural mismatch, not individual pathology, disrupts perception, attention, and decision-making, creating confusion and isolation. The public should care about this research because it reframes veterans' post-combat struggles as predictable consequences of cognitive architecture operating outside its natural ecological niche, highlighting how communities and institutions can design environments that better align with human cognition to support reintegration back home.

Research in cognitive science has revealed that the mind is built around primitive, automatic mechanisms rather than a holistic, single, general-purpose unit. In order to properly frame why veterans struggle with reintegration requires understanding what cognitive science means by architecture. Cognitive Architecture is the built-in, evolutionarily shaped systems that determine how information is processed. These systems are constrained by primitive information processing mechanisms. They are governed by a predisposed limited capacity buffer and automatic routines that mean they are not optional strategies that we can consciously control, but are instead the default settings of the human mind that work in tandem with the extrinsic environment to shape human nature. These constraints determine the lateral limits of how perception, attention, reasoning, and language work. In fact, as Cosmides and Tooby note, these systems are so adept and run so seamlessly that we often do not notice them unless they are

broken or misfire. This poses a set of problems for cognitive scientists, coined as instinct blindness. The field addresses this by utilizing Marr's Levels which takes a multi-step approach to understanding cognition. Marr posits that cognition cannot be fully understood from any one individual level or discipline; the three levels, which are computational, algorithmic, and implementation, must work together to approach accurate understanding. Utilizing Marr's approach, we can see that computational input, such as being consciously aware of being safe at a crowded grocery store, does not override the algorithmic processes that generate those threat signals. These same architectural constraints are recruited intensely in combat, where rapid inference and mandatory low-level processing become essential for survival.

Experiments and research on visual inference demonstrate how the brain solves the inverse problem by utilizing natural constraints instead of raw input. Attention and perception are two systems that modulate how we interact with the external world. Due to the inverse problem of vision, which is the attempt to portray a three-dimensional world projected onto a retina in two dimensions, our brain infers a lot about our environment. This results in vision being less like a live stream movie and more akin to a series of photographs, where what we perceive is dictated by where we attend in our visual field. The mind solves this by exploiting stable regularities in the world, but prolonged exposure to danger can tune higher-level systems so that ambiguous cues are interpreted as threatening. The low-level constraints do not change, but the thresholds and the weight of what is salient do. One such example of this is animacy detection and inferring agency; cognitive science experiments conducted in Yale's Perception and Cognition Lab have explored the psychophysics of chasing. They have found that subtle geometric clues, such as the orientation of one object towards another, can cause people to assign agency to the object and see it as something pursuing another object. This research is critical

because it highlights how what we thought of as high-level processes are happening automatically and making us perceive causal effects and social structures. With this in mind, we can see how these low-level systems that have been tuned for the chaos of combat can be hyperactive back home. A simple geometric shift of someone squaring their shoulders to you or the rapid change of someone's gait can trigger these systems to make the veteran feel like they are perceiving the intent of a would-be aggressor. This can cause a cascade of effects, a cortisol spike, an endocrine response, and attentional capture, where now the veteran consciously attends to possible threats in their environment, creating a feedback loop.

These loops are especially insidious because they cannot be broken with higher-level thought due to the modular nature of these systems. Cognitive science research into modularity has shown that the mind is not a homogenous unit. It is made up of specialized parts that are necessary for survival. Some processes are too valuable for survival to be slow; speed is of the essence, and as such, these modular systems are informationally encapsulated and have no access to external information. This makes these processes automatic and fast, but it also means that it involves mandatory processing. Understanding modularity helps us see how we fail in systematic and predictable ways. This insight is crucial to predicting the pitfalls associated with veterans reintegrating into society. While modularity cannot be erased or overridden, we can use the information gleaned from cognitive science research to understand that these modules are going to fire and be hypersensitive when returning home, and understanding this can help veterans notice when the feeling of danger does not match the actual situation. Recognizing this prediction error creates room for higher-level cognitive systems to slowly reshape how they interpret these automatic inputs with top-down inputs. Marr's Levels help clarify why this recalibration is slow; top-down beliefs function at the computation level, but the systems

generating these threat signals operate at the algorithmic level. The same modular design that makes threat perception automatic also shapes the reasoning system of how we handle social rules and expectations, and this system functions exceptionally well in the binary black and white structure of combat, but struggles in civilian life.

The mind is systematically bad at reasoning with abstract logic; we fail in consistent and predictable ways. From simple logical forms like failing to affirm the consequent or denying the antecedent, our mind seems poorly equipped to function in pure abstract logic that is separate from the social reasoning systems we evolved to handle. Research shows that people are naturally good at detecting rule violations in social settings, much better than abstract versions of that same detection. Combat is full of extremely rigid norms, with every service member having explicit responsibilities and a high penalty for any violations. The consequences of your actions immediately weigh on the group and can ultimately be a distinguishing factor between life and death. This environment mirrors the social setting that our reasoning systems evolved in, where social contract enforcement was both swift and necessary; as such, it is that much more jarring to go from an environment of explicit, rigid norms and expectations to a society with diffuse and ambiguously enforced behaviors. The dichotomy of these almost opposing systems can generate uncertainty, frustration, and isolation in the veteran. This isolation is further exacerbated by language and the connotations associated with in-group jargon.

Language is something we often take for granted because of how seamlessly we pick it up and utilize it in our day-to-day lives, but it is much more complex than initially believed. Cognitive scientists view language as something that is structured, modular, and automatic. It is built on innate genetic endowments that are refined with statistical learning. Fascinatingly enough, language does not only serve to convey ideas but can also function as a cipher.

Language and specific jargon can signal membership to a group, as seen in examples of shibboleths, denoted by shared vocabulary and structure. In the military, there is a plethora of in-group signals in the way that we communicate, from military jargon to clipped syntax, radio brevity, and acronyms. This shared language reinforces the feeling of community and further compounds the sense of unity servicemembers feel. Back home, the language that once held immense shared meaning is now rendered inert; there is no shared jargon, no instant recognition, no communion. This isolates the veteran socially in a way that is easily overlooked without the lens of language through a cognitive science perspective. The mismatch between these military and civilian settings weaves together to create a deepening chasm between the veteran and his civilian counterpart.

By this point, a clear theme is emerging: these systems make far more sense when viewed through evolution. This is because our cognitive architecture did not develop for modern civilian life; it is much better suited for ancestral environments, which are closer to combat than modern life is. The sheer urban density we face in modern life, as well as the anonymity, directly clash with the expectations of these systems. Therefore, it is especially jarring to go from a setting that feels tuned for this primitive architecture and then be plunged into environments that defy these expectations. From combat, where there are clear roles, a persistent threat, explicit group identity, and high-stakes coordination, to a foreign modern world that lacks all these elements. This leads to chronic stress, isolation, and attentional dysregulation, which carry both emotional and social consequences. Still, any explanation this broad has limits.

At the same time, it is important to acknowledge the limits of the framework itself. Of course, cognitive science is not a perfect nor complete model of the mind. Modularity is debated, and not every cognitive system has clear lateral limits, making it hard to know where one ends

and another begins. Culture and training also shape behavior in ways that evolutionary explanations alone cannot explain. But even with these limitations, the research still highlights something fundamental. Many of these systems veterans rely on in combat operate automatically and below conscious awareness. The mismatch is not absolute or predetermined, but it is strong enough to create predictable systematic struggles when these environments shift. However, these shortcomings do not weaken the framework; they reinforce why the architectural perspective helps us understand reintegration in a more realistic way, and as a far more complicated and involved process than we initially thought. This gap between what architecture requires and what institutions provide becomes clear when looking at actual transition practices. I spent nine years in the United States Marine Corps. The training process for my first deployment to Iraq took a year. Subsequent training for deployments to Afghanistan and Africa took 7-10 months each. When I left the Marine Corps, I received a weeklong course training to help with reintegration back to civilian life. Cognitive science demonstrates that this is insufficient, and it poorly equips transitioning veterans with the tools necessary to work against their very cognitive architecture.

Armed with this framework, we see that these everyday veteran struggles are not necessarily pathology but instead a mismatch between their cognitive architecture and the environment they are now reintegrating into. The direct result of systems that are mandatory, automatic, and informationally encapsulated. This reframes the entire conversation. These threat modules tuned for persistent danger will not simply turn off because a veteran knows they are home. The very systems that helped them to survive are now maladaptive to modern civilian life. This research is necessary because it shows us just how inadequate current transition models are. Cognitive science shows us that improving reintegration requires reshaping our approach, from



establishing shared language, creating predictable routines and social structures, and minimizing high-ambiguity environments in the early stages of transition.

Broadening the lens beyond individual veterans, this research has implications for the civilian public as well. The reason the public should care about this ongoing research is because it provides insight into the architectural constraints that influence veterans' behavior and reduces the overall stigma. This makes it easier to design support structures and influence policy within the Department of Defense and the VA. It changes the problem from an individual failing to a question of design. The more this research is brought to the public's attention, the more effectively we can understand and support those returning home. Our perception, inferences, and interpretations come from a set of modular, automatic, and informationally encapsulated systems, shaping our worldview long before our beliefs have entered the picture. These systems excel when immersed in an environment that mirrors the evolutionary pressures they evolved to address.

Ultimately, Conrad's claim that we live as we dream addresses something that feels fundamentally self-evident. He captures a persistent, uncomfortable, nagging truth that our individual experiences resist translation. Cognitive science helps us to understand why. We are alone not because our experiences are vastly different, unique, or ineffable. We are alone because our minds run on machinery that we cannot access, built for a world that no longer exists. Understanding that architecture is the first step towards designing reintegration systems that transform coming home from a solitary awakening into a shared return.